

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12397312
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	O'Brien; Kristen et al.

Machine motion cam track follower

Abstract

The present application provides a variable spray system for spraying a workpiece with a sprayable material at different angles. The variable spray system may include a linear actuator, a cam follower assembly, and a spray arm assembly. The linear actuator drives the cam follower assembly such that the cam follower assembly positions the spray arm assembly at a first angle with respect to the workpiece in a first stroke of the linear actuator and positions the spray arm assembly at a second angle with respect to the workpiece in a second stroke of the linear actuator.

Inventors: O'Brien; Kristen (Greenville, SC), Salm; Jacob A. (Greenville, SC), Sarratt; Craig (Greenville, SC)

Applicant: General Electric Company (Schenectady, NY)

Family ID: 1000008781993

Assignee: GE INFRASTRUCTURE TECHNOLOGY LLC (Greenville, SC)

Appl. No.: 18/150883

Filed: January 06, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240226945 A1	Jul. 11, 2024

Publication Classification

Int. Cl.: B05B15/68 (20180101); F16H25/06 (20060101)

U.S. Cl.:

CPC B05B15/68 (20180201); F16H25/06 (20130101);

Field of Classification Search

CPC: B05B (15/68); F16H (25/06)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2507640	12/1949	MacDonald	N/A	N/A
3696685	12/1971	Steele	N/A	N/A
5522706	12/1995	Mannava et al.	N/A	N/A
5846054	12/1997	Mannava et al.	N/A	N/A
6267558	12/2000	Dingwell et al.	N/A	N/A
7515547	12/2008	McCann et al.	N/A	N/A
7516547	12/2008	McCann et al.	N/A	N/A
9453576	12/2015	Schumann et al.	N/A	N/A
9856734	12/2017	Salm et al.	N/A	N/A
2006/0101883	12/2005	Garza	72/53	B24C 1/10
2008/0268757	12/2007	Mase et al.	N/A	N/A
2013/0180952	12/2012	Berkebile	216/41	C23C 10/04
2015/0031272	12/2014	Fulton et al.	N/A	N/A
2015/0298288	12/2014	Fulton et al.	N/A	N/A
2015/0375322	12/2014	Salm et al.	N/A	N/A
2016/0141200	12/2015	Takano	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
106140549	12/2015	CN	B05B 13/0228
0423041	12/1990	EP	B05B 13/0421
10-1381357	12/2013	KR	N/A
10-2113496	12/2019	KR	N/A

OTHER PUBLICATIONS

Machine translation of CN 106140549 A, obtained from FIT database (Year: 2016). cited by examiner
International Search Report and Written Opinion, PCT/US2023/086337, dated May 1, 2024 (8 pp.). cited by applicant

Primary Examiner: Diaz; Thomas C
Attorney, Agent or Firm: Eversheds Sutherland (US) LLP

Background/Summary

TECHNICAL FIELD

(1) The present application and the resultant patent relate generally to methods and systems for performing repetitive tasks and more particularly relate to methods and systems for rotational and translational movement of a tool such as a shot peening tool for a turbine engine rotor blade.

BACKGROUND

(2) Generally described, turbine engine rotor blades include an airfoil extending outwardly from a supporting platform and a dovetail. The dovetail is configured with axial lobes or tangs for mounting each blade in corresponding dovetail slots in the perimeter of a supporting rotor disk. During operation, energy is extracted from the hot combustion gases that flow past the turbine rotor blades which, in turn, rotate the supporting rotor disk for powering a compressor in a typical configuration. The rotor blades are subject to centrifugal loads during operation. These loads are carried radially inwardly through the supporting dovetails into the perimeter of the supporting rotor disk.

(3) The turbine rotor blades are typically formed of high strength superalloy materials having enhanced strength at the elevated temperatures typically found in the turbine. To further enhance the strength of the turbine rotor blades, the various surfaces of the dovetails may be shot peened in one of the last manufacturing steps producing the blades. Shot peening is a process in which metal shot is discharged in a stream of pressurized air over the surface of a metal workpiece to plastically deform the surface layer thereof and introduce residual compressive stress therein. The residual compressive stress reduces the stresses experienced in the component during operation, such as in the rotating environment of the gas turbine engine.

(4) Uniform shot peening of the entire turbine blade dovetail may provide maximum strength of the blade during operation and a correspondingly long service life. In order to ensure uniform coverage, many types of shot peening equipment use multiple nozzles positioned at varying angles. Each additional nozzle and the associated components, however, add to the overall cost, complexity, and efficiency of the equipment.

SUMMARY

(5) The present application and the resultant patent thus provide a variable spray system for spraying a workpiece with a spray-able material at different angles. The variable spray system may include a linear actuator, a cam follower assembly, and a spray arm assembly. The linear actuator drives the cam follower assembly such that the cam follower assembly positions the spray arm assembly at a first angle with respect to the workpiece in a first stroke of the linear actuator and positions the spray arm assembly at a second angle with respect to the workpiece in a second stroke of the linear actuator. The workpiece may be a turbine blade dovetail and the spray-able material may be shot peen.

(6) The present application and the resultant patent further provide a method of positioning a nozzle driven by a shaft of a cam follower assembly at different angles about a workpiece. The method may include the steps of driving the shaft in an upward direction, wherein driving the shaft in the upward direction causes the cam follower assembly to rotate the shaft and the nozzle in a first angle, and driving the shaft in a downward direction, wherein driving the shaft in the downward direction causes the cam follower assembly to rotate the shaft and the nozzle in a second angle.

(7) The present application and the resultant patent further provide a variable spray system for spraying a dovetail of a rotor blade with shot peen at different angles. The variable spray system may include a linear actuator, a cam follower assembly, and a spray arm with a nozzle. The linear actuator drives the cam follower assembly such that the cam follower assembly positions the spray arm and the nozzle at a first angle with respect to the dovetail in a first stroke of the linear actuator and positions the spray arm and the nozzle at a second angle with respect to the dovetail in a second stroke of the linear actuator.

(8) These and other features and improvements of this application and the resultant patent will become apparent to one of ordinary skill in the art upon review of the following detailed description when taken in conjunction with the several drawings and the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram of a gas turbine engine including a compressor, a combustor, a turbine, and an external load.
- (2) FIG. 2 is a perspective view of a turbine rotor blade.
- (3) FIG. 3 is a side view of a variable spray system as may be described herein.
- (4) FIG. 4 is a sectional view of the variable spray system of FIG. 3 taken along line 4-4.
- (5) FIG. 5 is a perspective view of a cam follower assembly of the variable spray system of FIG. 3.
- (6) FIG. 6 is a plan view of a cam track of the cam follower assembly of FIG. 5.
- (7) FIG. 7 is a schematic diagram of the spray nozzle of the variable spray system of FIG. 3 being positioned at different angles with respect to a workpiece.

DETAILED DESCRIPTION

(8) Referring now to the drawings, in which like numerals refer to like elements throughout the several views, FIG. 1 shows a schematic diagram of a gas turbine engine **10** as may be used herein. The gas turbine engine **10** may include a compressor **15**. The compressor **15** compresses an incoming flow of air **20**. The compressor **15** delivers the compressed flow of air **20** to a number of combustor cans **25**. The combustor cans **25** mix the compressed flow of air **20** with a pressurized flow of fuel **30** and ignite the mixture to create a flow of hot combustion gases **35**. Although only a single combustor can **25** is shown, the gas turbine engine **10** may include any number of combustor cans **25** positioned in a circumferential array and the like. Alternatively, the combustor **25** may be an annular combustor. The flow of combustion gases **35** is in turn delivered to a turbine **40**. The flow of combustion gases **35** drives the turbine **40** so as to produce mechanical work. The mechanical work produced in the turbine **40** drives the compressor **15** via a rotor shaft **45** and an external load **50** such as an electrical generator and the like.

(9) The gas turbine engine **10** may use natural gas, various types of syngas, liquid fuels, and/or other types of fuels and blends thereof. The gas turbine engine **10** may be any one of a number of different gas turbine engines offered by General Electric Company of Schenectady, New York, including, but not limited to, those such as a 7-series or a 9-series heavy duty gas turbine engine and the like and may be part of a simple cycle or a combined cycle power generation system. The gas turbine engine **10** may have different configurations and may use other types of components. Other types of gas turbine engines also may be used herein. Multiple gas turbine engines, other types of turbines, and other types of power generation equipment also may be used herein together.

(10) As described above, turbine engine rotor blades include an airfoil extending outwardly from a supporting platform and dovetail. To further enhance the strength of the turbine rotor blades, the various surfaces of the dovetails may be shot peened to reduce stresses therein. FIG. 2 shows a workpiece **55** in the form of a gas turbine engine rotor blade **60**. The rotor blade **60** includes an airfoil **65** extending outwardly from a supporting platform **70** with an integrally formed dovetail **75**. The dovetail **75** may be conventionally configured as an axial-entry dovetail with a number of serrations or dovetail lobes configured for mounting the rotor blade to the perimeter of a supporting rotor disk (not shown) having corresponding axial dovetail slots extending about the perimeter thereof. Other types of rotor blades **60** and the like also may be used herein.

(11) FIGS. 3-6 show an example of a variable spray system **100** as may be described herein. The variable spray system **100** may be used to shot peen a workpiece **55** such as the dovetail **75** of the rotor blade **60** and the like. Other types of workpieces **55** and other types of configurations may be

used herein. The variable spray system **100** may be in communication with a shot peen material source **110** with a volume of a shot peen material **120** therein. Other types of a spray-able materials **125** also may be used herein. The variable spray system **100** may be in communication with the shot peen material source **110** via a flexible hose **130** and the like.

(12) The variable spray system **100** includes a frame **140**. The frame **140** may have any suitable size, shape, or configuration. A cam follower assembly **150** may be mounted in the frame **140**. The cam follower assembly **150** maneuvers a spray arm assembly **160** via a linear actuator **170**. The linear actuator **170** may be of conventional design with a drive motor **180** attached to a linear motion shaft **190**. The linear motion shaft **190** may raise and lower an attached lift plate **200**. The length of travel of the lift plate **200** may be controlled by a limit switch (not shown) positioned within a T-slot mount **210** and the like. Other types of linear drive devices may be used herein. Other components and other configurations may be used herein.

(13) The cam follower assembly **150** includes a vertically driven shaft **220**. The shaft **220** may be positioned on the lift plate **200** of the linear actuator **170** for vertical movement therewith. A track roller **230** may be attached to the shaft **220** via a shaft coupling **240**. The shaft **220** and the track roller **230** may be positioned within a cam follower tube **250**. The cam follower tube **250** defines a track **260** therein. The track **260** may be largely in the shape of a parallelogram **270** with an upward vertical edge **280**, and upward angled edge **290**, a top recess **300**, a downward vertical edge **310**, a downward angled edge **320**, and a bottom recess **330**. The configuration of the track **260** and the edges therein may vary depending upon the desired resulting motion. Specifically, the track roller **230** is driven up and down the upward vertical edge **280** and the downward vertical edge **310** by the shaft **220** while the track roller **230** is driven at an angle along the upward angled edge **290** and the downward angled edge **320**. This angled motion causes the track roller **230** to rotate the shaft **220**. Other components and other configurations may be used herein.

(14) The spray arm assembly **160** may be attached to the shaft **220** for vertical and rotational movement therewith. In this example, the spray arm assembly **160** may include a horizontal arm **340**, an angled arm **350**, and a vertical arm **360**. The respective lengths of the arms **340**, **350**, **360** may vary depending on the desired resulting motion. The horizontal arm **340** may be attached to the shaft **220** for movement therewith. The vertical arm **360** may be attached to a spray nozzle **370**. The spray nozzle **370** may be in communication with the shot peen material source **110** via the flexible hose **130**. The spray nozzle **370** may be of conventional design. Other components and other configurations may be used herein.

(15) In use as is shown in FIG. 7, the variable spray system **100** may be used to spray the shot peen material **120** or other type of spray-able material **125** on to the rotor blade dovetail **75** or other types of workpieces **55**. Specifically, the variable spray system **100** may spray the shot peen material **120** from the bottom to the top of one side or at a first angle **380** of the dovetail **75** and then rotate to spray the shot peen material **120** from the top to the bottom of the other side or at a second angle **390** before rotating back to the starting position. In an upward or a first stroke **400**, the linear actuator **170** raises the shaft **220** as the track roller **230** travels along the upward vertical edge **280** of the track **260**. The linear actuator **170** continues to raise the shaft **220** as the track roller **230** travels along the upward angled edge **290** of the track **260** and the track roller **230** rotates the shaft **220** until the track roller **230** reaches the top recess **300**. In a downward or a second stroke **410**, the reverse occurs with the linear actuator **170** lowering the shaft **220** as the track roller **230** travels along the downward vertical edge **310** of the track. The linear actuator **170** continues to lower the shaft **220** as the track roller **230** travels along the downward angled edge **320** of the track **260** and the track roller **230** rotates the shaft **220** until the track roller **230** returns to the bottom recess **330**.

(16) This motion then may be repeated without any reset as the linear actuator **120** raises and lowers the shaft **220**. The reciprocating motion of the linear actuator **170** thus provides vertical and rotation movement to the shaft **220** and, hence, the spray arm assembly **160** and the nozzle **370**.

The nozzle **370** may make several passes on both sides or to both angles **380, 390** of the dovetail **75** as needed.

(17) The length of each stroke of the linear actuator **170**, the length of the vertical edges **280, 310** of the track **260**, the length and the angle of the angled edges **290, 320** of the track **260**, and the length of the arms **340, 350, 360** of the spray arm assembly **160** all may be varied individually or in combination to change the vertical distance traveled by the nozzle **370** as well as the degrees of rotation of the nozzle **370** about the dovetail **75** or other type of workpiece **55**.

(18) The variable spray system **100** thus utilizes the linear actuator **170** and the cam follower assembly **150** to create repetitive rotational actuation, thus simplifying the variable spray system **100** by reducing the number of components therein and, hence, eliminating waste. As described above, two nozzle generally were required for the shot peen process instead of the single nozzle **370** used herein. The combination of the linear actuator **170** and the cam follower assembly **150** may be used in any mechanical actuation application. For example, machines in a manufacturing environment requiring repetitive motion. This combination limits waste (energy, hardware, motion, controls) by utilizing a mechanical mechanism to create controlled rotational motion from linear motion.

(19) It should be apparent that the foregoing relates only to certain embodiments of this application and resultant patent. Numerous changes and modifications may be made herein by one of ordinary skill in the art without departing from the general spirit and scope of the invention as defined by the following claims and the equivalents thereof.

(20) Further aspects of the invention are provided by the subject matter of the following clauses: 1. A variable spray system for spraying a workpiece with a spray-able material at different angles, comprising: a linear actuator; a cam follower assembly; and a spray arm assembly; wherein the linear actuator drives the cam follower assembly such that the cam follower assembly positions the spray arm assembly at a first angle with respect to the workpiece in a first stroke of the linear actuator and positions the spray arm assembly at a second angle with respect to the workpiece in a second stroke of the linear actuator. 2. The variable spray system of clause 1, wherein the cam follower assembly comprises a shaft extending between the linear actuator and the spray arm assembly. 3. The variable system of any preceding clause, wherein the linear actuator comprises a lift plate and wherein the lift plate raises and lowers the shaft. 4. The variable spray system of any preceding clause, wherein the cam follower assembly comprises a track roller attached to the shaft. 5. The variable spray system of any preceding clause, wherein the cam follower assembly comprises a track and wherein the track roller travels along the track. 6. The variable spray system of any preceding clause, wherein the track comprises a parallelogram shape. 7. The variable spray system of any preceding clause, wherein the track comprises an upward vertical edge, an upward angled edge, a downward vertical edge, and a downward angled edge. 8. The variable spray system of any preceding clause, wherein the track roller travels along the upward vertical edge and the upward angled edge of the track during the first stroke of the linear actuator. 9. The variable spray system of any preceding clause, wherein the track roller causes the shaft to rotate while the track roller travels along the upward angled edge of the track. 10. The variable spray system of any preceding clause, wherein the track roller travels along the downward vertical edge and the downward angled edge of the track during the second stroke of the linear actuator. 11. The variable spray system of any preceding clause, wherein the track roller causes the shaft to rotate while the track roller travels along the downward angled edge of the track. 12. The variable spray system of any preceding clause, wherein the spray arm assembly comprises a nozzle. 13. The variable spray system of any preceding clause, wherein the workpiece comprises a dovetail of a rotor blade. 14. The variable spray system of any preceding clause, wherein the spray-material comprises a shot peen material. 15. A method of positioning a nozzle driven by a shaft of a cam follower assembly at different angles about a workpiece, comprising: driving the shaft in an upward direction; wherein driving the shaft in the upward direction causes the cam follower assembly to rotate the shaft in a

first angle; and driving the shaft in a downward direction; wherein driving the shaft in the downward direction causes the cam follower assembly to rotate the shaft in a second angle. 16. A variable spray system for spraying a dovetail of a rotor blade with shot peen at different angles, comprising: a linear actuator; a cam follower assembly; and a spray arm with a nozzle; wherein the linear actuator drives the cam follower assembly such that the cam follower assembly positions the spray arm and the nozzle at a first angle with respect to the dovetail in a first stroke of the linear actuator and positions the spray arm and the nozzle at a second angle with respect to the dovetail in a second stroke of the linear actuator. 17. The variable spray system of any preceding clause, wherein the cam follower assembly comprises a shaft extending between the linear actuator and the spray arm. 18. The variable spray system of any preceding clause, wherein the cam follower assembly comprises a track roller attached to the shaft and wherein the cam follower assembly comprises a track. 19. The variable spray system of any preceding clause, wherein the track comprises an upward vertical edge, an upward angled edge, a downward vertical edge, and a downward angled edge. 20. The variable spray system of any preceding clause, wherein the track roller causes the shaft to rotate in the first angle while the track roller travels along the upward angled edge of the track and wherein the track roller causes the shaft to rotate in the second angle while the track roller travels along the downward angled edge of the track.

Claims

1. A variable spray system for spraying a workpiece with a spray-able material at different angles, comprising: a vertically oriented linear actuator; a cam follower assembly; and a spray arm assembly; wherein the vertically oriented linear actuator drives the cam follower assembly such that the cam follower assembly positions the spray arm assembly at a first angle and a first height with respect to the workpiece in a first stroke of the vertically oriented linear actuator and positions the spray arm assembly at a second angle and a second height with respect to the workpiece in a second stroke of the vertically oriented linear actuator.
2. The variable spray system of claim 1, wherein the cam follower assembly comprises a shaft extending between the vertically oriented linear actuator and the spray arm assembly.
3. The variable spray system of claim 2, wherein the vertically oriented linear actuator comprises a lift plate and wherein the lift plate raises and lowers the shaft.
4. The variable spray system of claim 2, wherein the cam follower assembly comprises a track roller attached to the shaft.
5. The variable spray system of claim 4, wherein the cam follower assembly comprises a track and wherein the track roller travels along the track.
6. The variable spray system of claim 5, wherein the track comprises a parallelogram shape with a pair of parallel vertical edges and a pair of parallel angled edges.
7. The variable spray system of claim 5, wherein the track comprises an upward vertical edge, an upward angled edge, a downward vertical edge, and a downward angled edge.
8. The variable spray system of claim 7, wherein the track roller travels along the upward vertical edge and the upward angled edge of the track during the first stroke of the vertically oriented linear actuator.
9. The variable spray system of claim 8, wherein the track roller causes the shaft to rotate while the track roller travels along the upward angled edge of the track.
10. The variable spray system of claim 7, wherein the track roller travels along the downward vertical edge and the downward angled edge of the track during the second stroke of the vertically oriented linear actuator.
11. The variable spray system of claim 10, wherein the track roller causes the shaft to rotate while the track roller travels along the downward angled edge of the track.
12. The variable spray system of claim 1, wherein the spray arm assembly comprises a nozzle.

13. The variable spray system of claim 1, wherein the workpiece comprises a dovetail of a rotor blade.

14. The variable spray system of claim 1, wherein the spray-able material comprises a shot peen material.

15. A method of positioning a nozzle driven by a shaft of a cam follower assembly at different angles about a workpiece, comprising: driving the shaft in an upward direction via a vertically oriented linear actuator; wherein driving the shaft in the upward direction causes the cam follower assembly to rotate the shaft in a first angle and raise the nozzle to a first height; and driving the shaft in a downward direction via the vertically oriented linear actuator; wherein driving the shaft in the downward direction causes the cam follower assembly to rotate the shaft in a second angle and lower the nozzle to a second height.

16. A variable spray system for spraying a dovetail of a rotor blade with shot peen at different angles, comprising: a vertically oriented linear actuator; a cam follower assembly; and a spray arm with a nozzle; wherein the vertically oriented linear actuator drives the cam follower assembly such that the cam follower assembly positions the spray arm and the nozzle at a first height and a first angle with respect to the dovetail in a first stroke of the vertically oriented linear actuator and positions the spray arm and the nozzle at a second height and a second angle with respect to the dovetail in a second stroke of the vertically oriented linear actuator.

17. The variable spray system of claim 16, wherein the cam follower assembly comprises a shaft extending between the vertically oriented linear actuator and the spray arm.

18. The variable spray system of claim 17, wherein the cam follower assembly comprises a track roller attached to the shaft and wherein the cam follower assembly comprises a track.

19. The variable spray system of claim 18, wherein the track comprises an upward vertical edge, an upward angled edge, a downward vertical edge, and a downward angled edge.

20. The variable spray system of claim 19, wherein the track roller causes the shaft to rotate in the first angle while the track roller travels along the upward angled edge of the track and wherein the track roller causes the shaft to rotate in the second angle while the track roller travels along the downward angled edge of the track.
